



Zero-Shot and Few-Shot Generalization for Novel Image Editing Tasks using Aligned Foundation Models

Domeniu: Informatică sau CTI

Descriere:

Our work

We are developing AI-driven editing systems that push beyond narrow, task-specific pipelines toward broadly generalizable models. By leveraging large-scale foundation models—such as Flux, Firefly, NanoBanana and Stable Diffusion—we aim to enable users to perform complex or personalized image edits without requiring extensive retraining or annotated data. Through lightweight adaptation strategies, multimodal alignment, and transfer learning, we are making creative AI tools more flexible, accessible, and scalable across diverse editing tasks.

What we offer

Work on advancing zero-shot and few-shot learning approaches for image editing with direct applicability to next-generation creative platforms. Your research will contribute to building AI systems capable of robust generalization across new tasks with minimal supervision. Collaborate with research and engineering teams to explore model alignment, adaptation, and benchmarking, ensuring that results move seamlessly from experimental design into practical, real-world solutions.

Project Summary

This project investigates methods for enabling zero-shot and few-shot generalization in image editing tasks by aligning vision-language foundation models with generative diffusion models. Unlike conventional fine-tuning approaches that require task-specific data, this research focuses on conditioning models through task descriptions, examples, or synthetic demonstrations to enable flexible, data-efficient editing. Core research questions include:

- How can task descriptions or examples (without explicit retraining) be used to condition generative editing pipelines?

- How can vision-language representations be aligned with diffusion models for robust transfer learning?
- What are the practical limits of zero-shot editing, and how can few-shot methods bridge the gap?

Potential contributions include:

- New strategies for zero-shot task conditioning (e.g., text-only prompts, image-text pairs, synthetic examples);
- A framework for combining frozen foundation models with lightweight adapters or prompt-based tuning (e.g., LoRA, DreamBooth);
- A benchmark dataset or protocol for evaluating generalization across diverse editing types.

This research aims to expand the adaptability and scalability of AI-driven editing systems, enabling personalized, task-specific functionality even in settings with little or no training data.